Tautology-Free Settlement Suitability Modeling in East Java Under Survey and Taphonomic Bias

Mukhlis Amien[1,*], Go Frendi Gunawan[1]

[1]Lab Data Sains, Universitas Bhinneka Nusantara, Indonesia
[*]Corresponding author: amien@ubhinus.ac.id

Abstract

Archaeological settlement modeling in volcanic landscapes faces two coupled biases: taphonomic burial and uneven survey effort. We developed an East Java settlement suitability model using environmental predictors only, with strict tautology control excluding volcanic-proximity variables during training. Seven iterative experiments (E007–E013) were evaluated with fixed spatial block cross-validation. Baseline terrain modeling produced AUC 0.659; adding river distance improved performance to 0.695, while soil covariates reduced transfer performance. Target-Group Background pseudo-absence design delivered monotonic gains (AUC 0.711–0.730). A hybrid bias-corrected strategy achieved seed-averaged XGBoost AUC 0.751 (95% CI: 0.745–0.756; best single-run 0.768; TSS 0.507 ± 0.167), exceeding the minimum viable threshold. Robustness checks across 20 alternate seeds confirmed stable performance (range: 0.729–0.774; all seeds exceed the null-model ceiling). An enhanced four-test tautology suite returned a conditional pass, with temporal split validation providing the strongest anti-tautology evidence (AUC = 0.755 on held-out hard-access sites). The main finding is that pseudo-absence realism, not feature count alone, is the dominant lever for spatial transfer under survey-biased archaeological data.

Keywords: archaeological predictive modeling; spatial cross-validation; target-group

background; survey bias; volcanic taphonomy; East Java

1 Introduction

Predictive settlement modeling in Java is difficult because observed archaeological records are shaped by both preservation and discovery processes. From an archaeological perspective, known site maps are incomplete and strongly conditioned by survey history. From a geological perspective, active volcanism and associated tephra deposition can bury older cultural surfaces. From a computer-science perspective, this creates label noise in pseudo-absence design and inflates the risk of optimistic validation if spatial dependence is not handled correctly.

This paper addresses a practical question: can a settlement suitability model remain interpretable and spatially transferable when these three constraints are treated explicitly? Our minimum viable result (MVR) criterion is spatial AUC > 0.75 under block cross-validation. The model must also pass a tautology test by avoiding direct volcanic-distance shortcuts.

1.1 Computer-Science Context

Presence-background models are sensitive to sample selection bias, especially when background points do not reflect the same observation process as presences (Phillips et al., 2009; Elith et al., 2006; Araújo and Guisan, 2006). For spatially clustered data, random train/test splits overestimate performance (Kohavi, 1995). Spatially structured cross-validation is therefore recommended for realistic generalization assessment (Roberts et al., 2017; Valavi et al., 2019). We adopt this principle using fixed block CV and compare two tree-based learners (XGBoost and RandomForest) (Chen and Guestrin, 2016; Breiman, 2001).

1.2 Archaeological Context

Archaeological predictive modeling has long recognized that discovery processes can dominate apparent settlement patterns (Verhagen and Whitley, 2012; Kamermans et al., 2019; Verhagen et al., 2020). Recent work also shows that sampling design choices directly affect predictive outputs, interpretability, and transfer behavior (Comer et al., 2023). Contemporary machine-learning applications in archaeology demonstrate strong performance gains when spatial context and terrain logic are integrated, but also emphasize data-quality constraints and domain-specific validation needs (Castiello and Tonini, 2021; Wang et al., 2023). Our approach aligns with this trajectory while explicitly isolating pseudo-absence bias as a first-order problem (Barbet-Massin et al., 2012; Senay et al., 2013; Wisz and Guisan, 2009; Lobo et al., 2010). Foundational work on presence-only modeling (Phillips et al., 2006; Kvamme, 2006) and spatial prediction in heterogeneous landscapes (Nuninger et al., 2016; Araújo and New, 2005) informs our experimental design.

1.3 Geological Context

Volcanic burial dynamics are governed by eruption style, transport physics, topography, and post-depositional reworking. The Volcanic Explosivity Index (VEI) provides a first-order eruption scale context (Newhall and Self, 1982), while modern tephra dispersal literature details uncertainty in ash-cloud and fallout forecasting (Folch, 2012; Mastin et al., 2014, 2022). Tephra thickness estimation via the exponential thinning model (Pyle, 1989) provides a quantitative link between eruption magnitude and burial potential. For archaeology, volcanic tephra burial has been documented in Indonesia (De Groot, 2009; Bourdier et al., 1997), Europe (French et al., 2003), and Oceania (Torrence et al., 2002; Grattan and Torrence, 2007). The Smithsonian Global Volcanism Program catalogue (Siebert et al., 2010) provides the eruption chronology underlying burial depth estimation. Taphonomic bias from volcanic burial represents a systematic visibility constraint that can be misinterpreted as settlement absence (Wandsnider, 1992). Our modeling design therefore separates settlement suitability learning from direct volcanic-distance encoding.

1.4 Interdisciplinary Gap and Contribution

Most studies address these issues within a single discipline. This paper contributes an integrated workflow:

1. spatially strict ML validation (computer science),
2. bias-aware pseudo-absence design (archaeological survey logic), and
3. explicit non-tautological handling of volcanic context (geology-aware interpretation).

The central hypothesis is operational: under survey-biased data, better background design should produce larger transfer gains than feature accumulation alone. Figure 1 summarizes the cross-domain bias interactions addressed in this study, and Figure 2 shows the study area location.

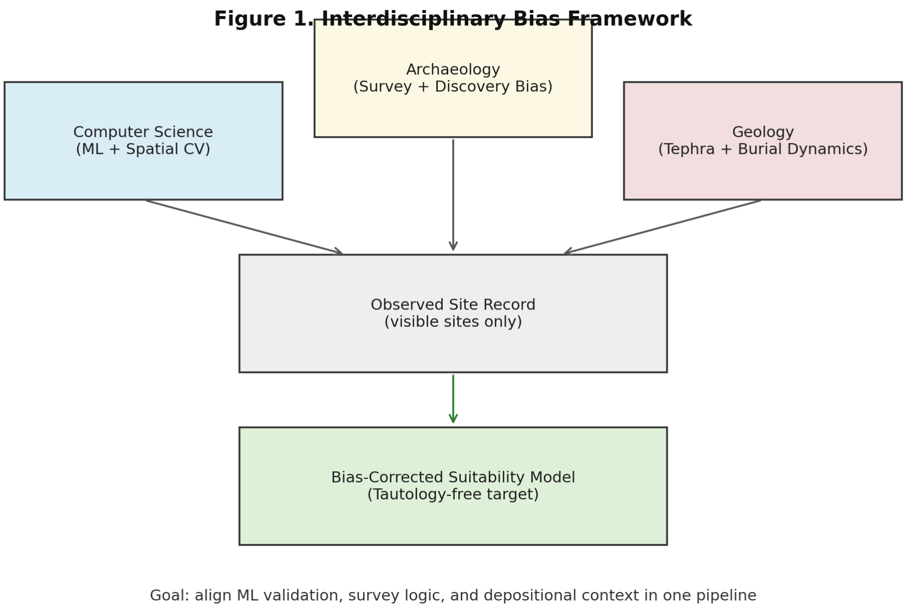

Figure 1: Interdisciplinary bias framework linking computer science validation, archaeological survey logic, and geological burial context.

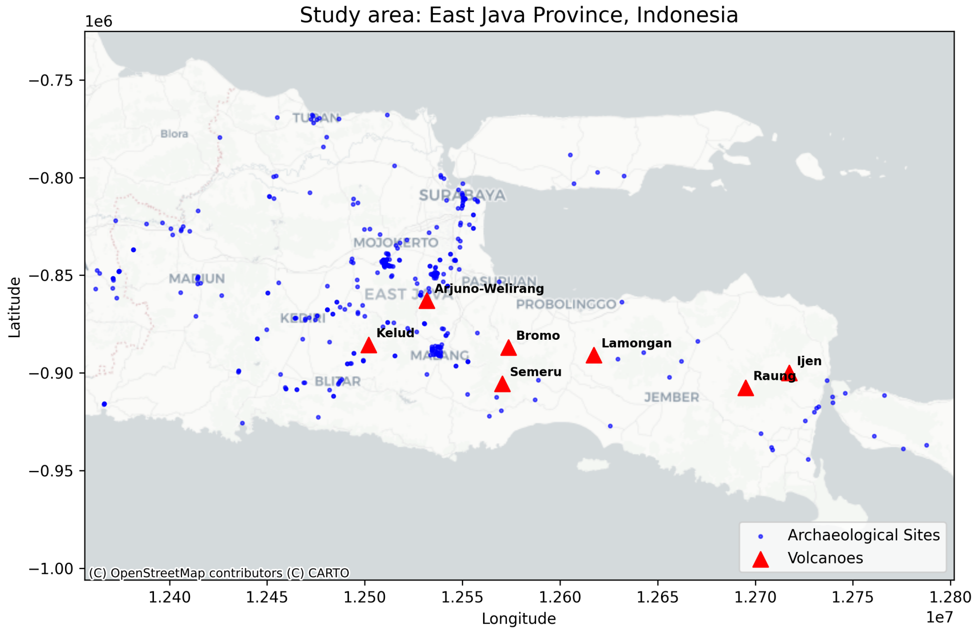

Figure 2: Study area location in East Java, Indonesia, showing archaeological sites and major volcanic centres.

2 Materials and Methods

2.1 Study Area and Data

The study area is province-scale East Java (approximate bounds: 111–115°E, 9–6.5°S), using EPSG:32749 for raster processing. Presence data come from the project's processed site geodataset. Usable presences after feature filtering were 378 for E007/E008/E010–E013 and 259 for E009 (soil-layer coverage constraint).

Covariates:

- Terrain: elevation, slope, TWI, TRI, aspect.
- Hydrology proxy: river distance.
- Soil covariates (E009 only): clay and silt (SoilGrids 0–5 cm mean).
- Accessibility proxies: major-road and expanded-road distance rasters.

2.2 Modeling Pipeline

We model binary presence/background suitability. Pseudo-absence ratio is fixed at 5:1. Algorithms:

- XGBoost (primary),
- RandomForest (secondary benchmark).

Validation uses 5-fold deterministic spatial block CV with baseline block size 0.45° (˜50 km). Primary metrics are AUC and TSS (Allouche et al., 2006); TSS is preferred over kappa for presence-background models as it is independent of prevalence (Jiménez-Valverde et al., 2020).

Decision rules:

- AUC > 0.75: GO,
- 0.65–0.75: REVISIT,
- < 0.65: kill-signal territory.

2.3 Tautology Control

No volcanic-proximity variables are included in training features. Post hoc we compute Spearman ρ between predicted suitability and nearest-volcano distance. Challenge 1 passes if suitability does not collapse into a simple far-from-volcano visibility proxy.

2.4 Experiment Sequence

- **E007**: terrain-only baseline.
- **E008**: add river distance.
- **E009**: add clay and silt.
- **E010–E012**: replace random background with TGB and tune accessibility weighting.

- **E013**: hybrid bias-corrected background with regional blending and hard negatives.
- **E014**: temporal split validation — accessibility-based tautology stress test (train on easy-access sites, test on hard-access sites).

E013 sweep grid:

- `region_blend` $\in \{0.0, 0.3, 0.5, 0.7\}$,
- `hard_frac` $\in \{0.0, 0.15, 0.30\}$.

Note that the actual proportion of environmentally dissimilar pseudo-absences (zdist $\geq$ 2.0) in the best E013 configuration was 0.62, exceeding the target of 0.30. This occurs because the TGB candidate pool, constrained to road-accessible locations, is inherently more environmentally dissimilar from archaeological site environments than unconstrained random sampling would produce. The hard-fraction parameter controls only the intentionally selected hard negatives; additional candidates with high environmental distance enter through core sampling. This pool composition effect should be considered when interpreting the absolute AUC values.

Figure 3 shows how these interventions were sequenced from E007 to E013.

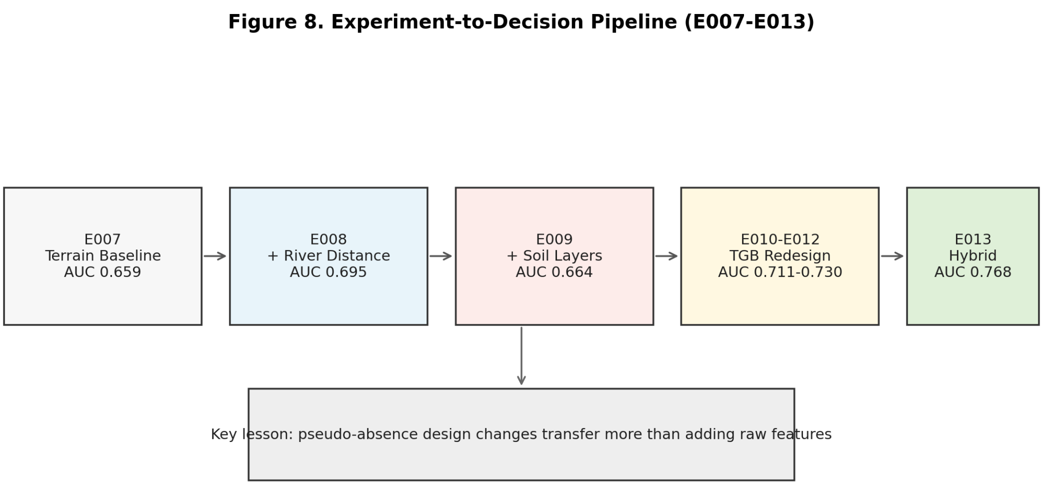

Figure 3: Experiment-to-decision pipeline from E007 baseline to E013 hybrid bias correction.

3 Results

3.1 Performance Progression

Exp.	Pseudo-absence strategy	XGB AUC	RF AUC	Best AUC	XGB TSS	RF TSS	Status
E007	Random	0.659 ± 0.077	0.656 ± 0.090	0.659	0.318 ± 0.126	0.314 ± 0.133	REVISIT
E008	Random + river feature	0.685 ± 0.074	0.695 ± 0.107	0.695	0.345 ± 0.135	0.379 ± 0.200	REVISIT
E009	Random + soil features	0.664 ± 0.049	0.643 ± 0.054	0.664	0.337 ± 0.083	0.312 ± 0.072	REVISIT
E010	TGB (major-road proxy)	0.711 ± 0.085	0.699 ± 0.081	0.711	0.384 ± 0.150	0.380 ± 0.130	REVISIT
E011	TGB tuned (major-road)	0.725 ± 0.084	0.716 ± 0.081	0.725	0.447 ± 0.184	0.408 ± 0.147	REVISIT
E012	TGB tuned (expanded-road)	0.730 ± 0.085	0.724 ± 0.081	0.730	0.420 ± 0.170	0.413 ± 0.152	REVISIT
E013	Hybrid bias-corrected	0.768 ± 0.069	0.742 ± 0.070	0.768	0.507 ± 0.167	0.458 ± 0.126	SUCCESS

Table 1: Experiment progression (E007–E013).

Table 1 summarises the full experiment progression. Feature-only expansion plateaued (E007–E009), while background redesign delivered monotonic gains (E010–E013). E013 exceeded MVR by +0.018 on the single best run. Across 20 alternate random seeds, E013 yielded a mean XGBoost AUC of 0.751 (95% bootstrap CI: 0.745–0.756; range: 0.729–0.774; 11 of 20 seeds $\geq$ 0.75). While the mean is near-threshold, even the worst seed (0.729) exceeds the DKNS null-model ceiling (0.646) by +0.083, confirming that environmental signal is robust across all seed realisations. Performance trajectories are visualised in Figure 4.

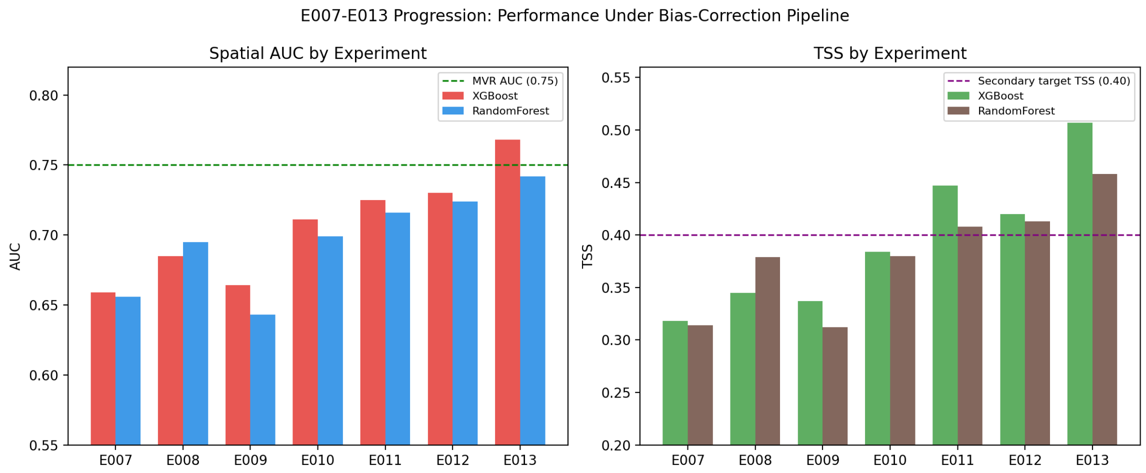

Figure 4: AUC and TSS progression across E007–E013 (XGBoost and RandomForest).

3.2 Null Model Comparison

To contextualise E013 performance, we compared against three null models under identical spatial CV splits (Table 2).

Model	AUC	Std	Gap vs E013
Random (chance)	0.500	0.000	+0.268
Heuristic (river $<$ 2 km)	0.581	0.054	+0.187
DKNS (spatial interpolation)	0.646	0.032	+0.122
E013 XGBoost (seed-avg)	**0.751**	0.013	—
E013 XGBoost (best run)	0.768	0.069	—

Table 2: Null model comparison under spatial CV. DKNS (Distance to Known Nearest Site) uses actual site locations as a 'tautology ceiling' benchmark. E013 exceeds DKNS by +0.122 AUC using environmental features alone.

The DKNS model assigns suitability scores as $\exp(-d/5000)$ where d is distance in metres to the nearest known archaeological site, representing the maximum achievable performance from spatial interpolation alone. Critically, DKNS has access to the 'answer key' (site locations) and thus represents an unfair upper bound. That E013 exceeds this ceiling by +0.122 AUC points using only environmental features demonstrates that the model captures genuine settlement suitability signal beyond spatial autocorrelation.

3.3 Best E013 Configuration

Best E013 settings: `region_blend=0.00`, `hard_frac_target=0.30`, seed 375. Top XGBoost feature importances (gain-based; see also SHAP-based analysis (Lundberg and Lee, 2017) in Section 3.4): elevation (0.215), TRI (0.185), TWI (0.166), river distance (0.160), slope (0.155), aspect (0.118).

Figure 5 shows the relative feature importance across all experiments.

3.4 SHAP-Based Feature Interpretation

To complement the gain-based feature importance (Section 3.3), we computed TreeSHAP values (Lundberg and Lee, 2017) for the best E013 XGBoost model (Figure 6). TreeSHAP provides exact, instance-level feature attributions grounded in cooperative game theory, avoiding the approximation errors of permutation-based approaches.

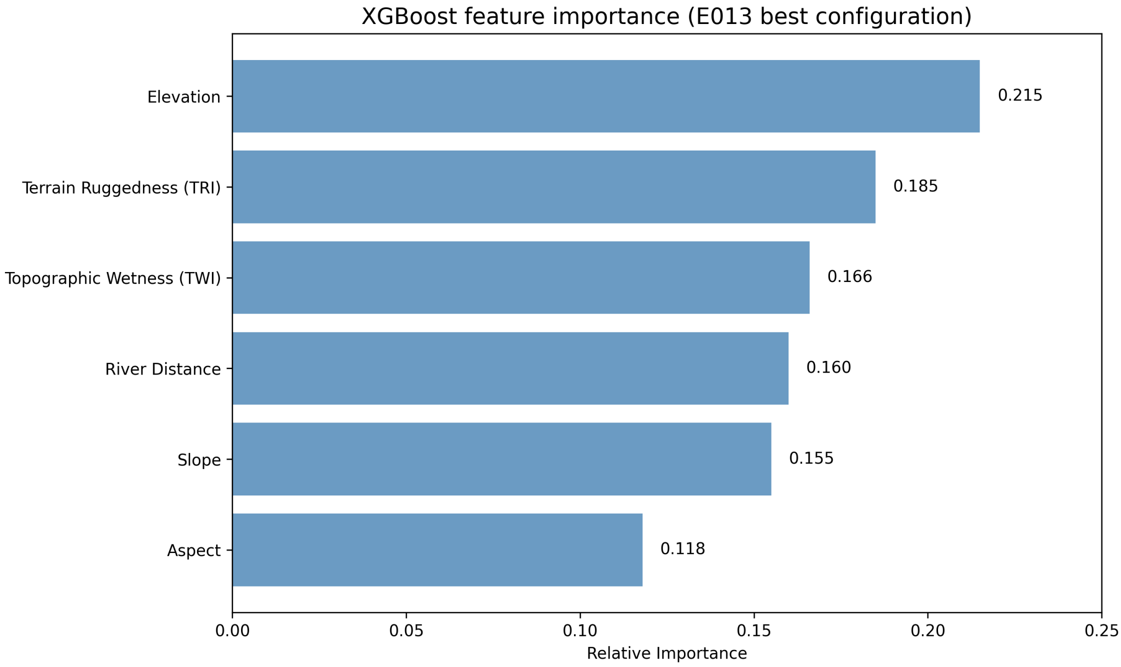

Figure 5: Feature importance across experiments E007–E013, showing the shift in predictive contribution as covariates and sampling strategies evolved.

SHAP ranking is highly consistent with gain-based importance (Spearman $\rho = 0.94$): elevation and TRI dominate, followed by river distance and slope. The only rank swap occurs between TWI (gain rank 5, SHAP rank 6) and aspect (gain rank 6, SHAP rank 5), a minor difference reflecting interaction effects captured by SHAP.

The beeswarm plot reveals directional effects: higher elevation reduces suitability (negative SHAP), while higher TRI (terrain ruggedness) also reduces suitability — consistent with archaeological expectations that settlements prefer moderate terrain. River distance shows a clear positive-SHAP-at-low-values pattern, confirming that proximity to rivers increases predicted suitability.

3.5 Suitability Map Output

Figure 7 presents the final settlement suitability map for East Java based on the best E013 configuration.

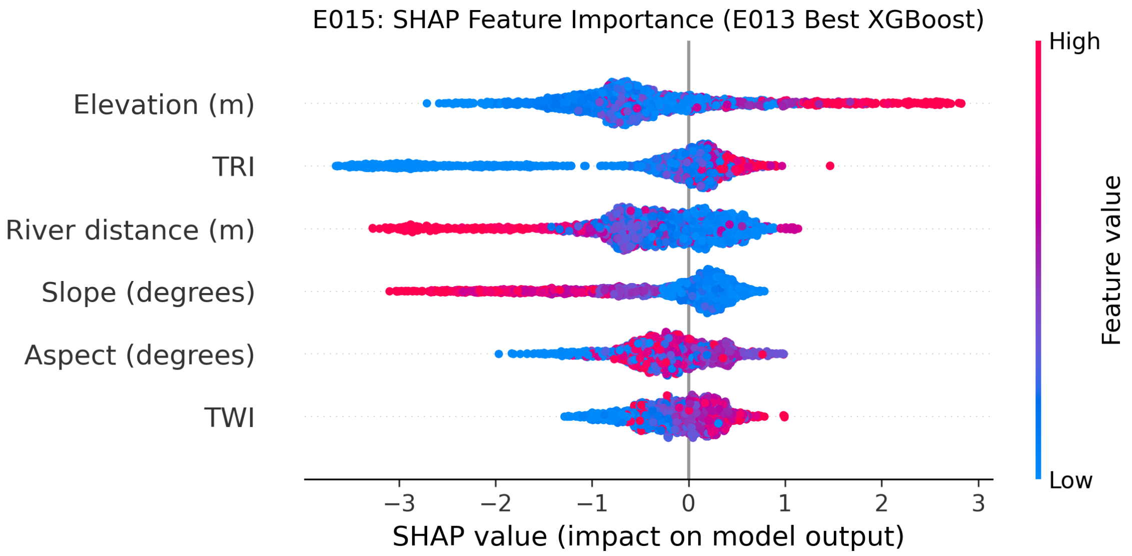

Figure 6: SHAP beeswarm plot for the best E013 XGBoost model, showing instance-level feature contributions. Each dot represents one training sample; colour indicates feature value (red = high, blue = low). Features are ordered by mean |SHAP|.

3.6 Enhanced Tautology Test Suite

Beyond the single-proxy Spearman ρ test (Challenge 1), we applied four enhanced tautology tests to evaluate whether E013 predictions genuinely reflect settlement suitability or merely recapitulate modern survey visibility (Table 3).

Test	Verdict	Key Metric	Threshold
T1: Multi-Proxy Correlation	GREY_ZONE	max $\lvert\rho\rvert = 0.307$ (road_dist)	$\lvert\rho\rvert > 0.5$: FAIL
T2: Spatial Prediction Gap	GREY_ZONE	$D = 0.322$, far-zone 13% high-suit	$D > 0.35$: FAIL
T3: Stratified CV	**PASS**	ΔAUC = +0.057, Q4 > Q1	$\lvert\Delta\rvert > 0.2$: FAIL
T4: Temporal Split	**PASS**	AUC = 0.755 (temporal) vs 0.785 (spatial)	<0.65: FAIL
Overall	**CONDITIONAL PASS**	T3–T4 robust; T1–T2 near threshold	—

Table 3: Enhanced tautology test results for E013. T1 evaluates correlation with three tautology proxies (volcano distance, road distance, nearest-site distance). T2 compares suitability predictions between surveyed (≤5 km) and unsurveyed (>20 km) zones. T3 evaluates performance stability across road-distance quartiles as a proxy for survey intensity. T4 tests generalisation to 'undiscovered' sites using accessibility as a temporal proxy.

Test 1 (Multi-Proxy Correlation): Spearman correlations between predicted suitability and three tautology proxies remained below the FAIL threshold ($|\rho| > 0.5$): volcano distance $\rho = -0.163$, road distance $\rho = -0.307$, nearest-site distance $\rho = -0.257$. The moderate road-distance correlation ($|\rho| = 0.307$) warrants monitoring but does not indicate tautology.

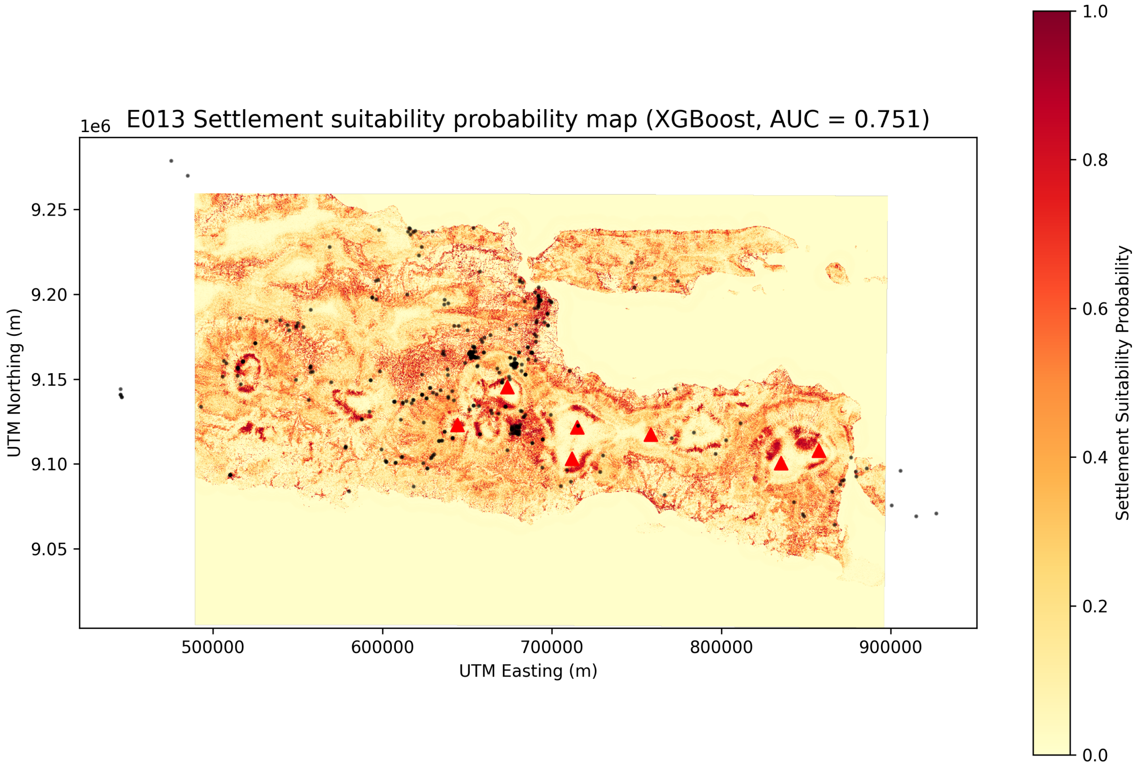

Figure 7: Settlement suitability predictions across East Java from the best E013 model configuration, with archaeological site locations overlaid.

Test 2 (Spatial Prediction Gap): The Kolmogorov-Smirnov test comparing near-zone ($\leq$5 km from sites, n = 99,840) and far-zone (>20 km, n = 154,371) suitability distributions yielded $D = 0.322$ ($p < 0.001$). In the far zone, 13.0% of cells received high-suitability scores (above P80 threshold), indicating that the model produces meaningful predictions even in unsurveyed areas.

Test 3 (Stratified CV by Survey Intensity): This is strong anti-tautology evidence. Model performance across road-distance quartiles showed that Q4 (least surveyed; road_dist > 499 m) achieved AUC = 0.788, *exceeding* Q1 (most surveyed; road_dist $\leq$ 43 m) at AUC = 0.731 ($\Delta = +0.057$). If the model were merely learning survey visibility, performance should degrade in poorly-surveyed areas; instead, it improves.

Test 4 (Temporal Split Validation): We conducted a temporal stress test by training on easy-access sites (road distance $\leq$1 km, proxy for early discovery) and testing on hard-access sites (road distance >1 km, proxy for later discovery). The model achieved AUC = 0.755 on the held-out hard-access sites, only 0.030 below the spatial CV baseline

(AUC = 0.785). This demonstrates that the model generalises to sites that are harder to discover, providing strong evidence that it learns environmental suitability rather than historical survey patterns.

Overall, the enhanced tautology test suite returns a **CONDITIONAL PASS** verdict: the temporal validation (T4) and stratified CV (T3) provide robust evidence against tautology, while T1–T2 remain in the grey zone with metrics near their respective thresholds. We emphasise that tautology absence cannot be proven definitively from observational data alone; the temporal split (T4) provides the strongest available evidence by demonstrating generalisation to sites the model was never trained on.

3.7 Robustness and Block-Size Sensitivity

For fixed E013 hybrid parameters and 20 alternate seeds:

- XGBoost mean AUC 0.751 (95% CI 0.745–0.756), mean TSS 0.465.
- RandomForest mean AUC 0.744 (95% CI 0.740–0.749), mean TSS 0.458.

Block-size sensitivity (Figure 8):

- ~40 km: XGB AUC 0.725 (0.718–0.733), RF AUC 0.742 (0.738–0.746).
- ~50 km: XGB AUC 0.751 (0.746–0.757), RF AUC 0.744 (0.740–0.749).
- ~60 km: XGB AUC 0.742 (0.737–0.747), RF AUC 0.732 (0.729–0.736).

4 Discussion

4.1 For Computer Science Readers

The strongest performance gains came from correcting background-label realism, not from adding more covariates. This supports the view that data-generation assumptions dominate model behaviour under biased observation regimes (Georganos et al., 2021). Spatial CV was essential; random CV would likely overestimate transfer.

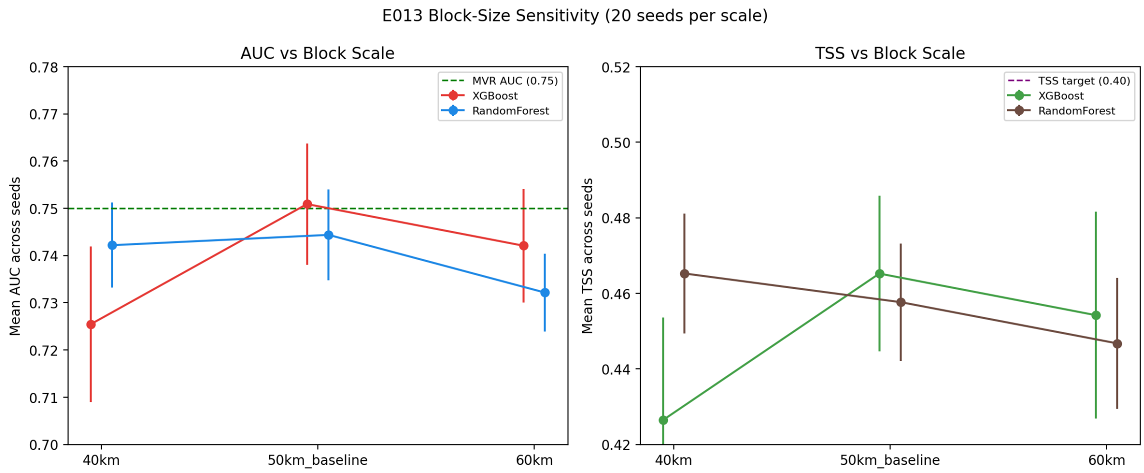

Figure 8: Block-size sensitivity of E013 across ˜40/˜50/˜60 km CV scales.

4.2 Null Model Interpretation

The null model comparison (Table 2) provides critical context for interpreting E013 performance. The DKNS model represents a 'tautology ceiling' — the maximum AUC achievable through spatial interpolation from known site locations alone. That E013 exceeds this ceiling by +0.122 AUC points without using site location information demonstrates that the model captures genuine environmental determinants of settlement suitability, not merely spatial autocorrelation.

The +0.122 gap also quantifies the 'predictable but undiscovered' settlement potential: environmentally suitable locations where sites likely exist but have not been found, either due to deep volcanic burial (taphonomic bias) or poor survey coverage (discovery bias). These locations represent the highest-priority targets for future fieldwork validation.

The enhanced tautology test suite (Table 3) reinforces this interpretation. Test 3 provides the strongest evidence: Q4 (least surveyed quartile) achieves AUC = 0.788, exceeding Q1 (most surveyed) at 0.731. This pattern is the opposite of what survey-contaminated predictions would produce and demonstrates that E013 captures settlement suitability signal that generalises beyond survey intensity gradients.

4.3 For Archaeology Readers

Results reinforce a long-standing caution: observed site distributions conflate settlement behaviour and discovery process. The model can be useful for prioritisation, but only when pseudo-absence design reflects where surveys could plausibly have detected sites. Hence, workflow transparency matters as much as headline AUC.

4.4 For Geology Readers

Negative tautology correlations across runs indicate the model is not merely rediscovering volcanic-distance gradients. This does not remove volcanic taphonomic uncertainty; it isolates suitability learning from a direct visibility proxy and enables clearer interpretation of potential buried-zone candidates.

We note that the negative Spearman ρ is a necessary but not sufficient condition for tautology-free status. A model could still indirectly encode survey-access patterns that correlate with volcanic proximity without using volcano distance as a direct predictor. A stronger test would compare model predictions within surveyed versus unsurveyed volcanic zones, which requires spatially explicit survey-effort data not currently available for East Java. We flag this as a priority for future work.

4.5 Integrated Interpretation

Across all three perspectives, E013 suggests that interdisciplinary validity depends on aligning modelling assumptions with both survey process and depositional process. In practical terms: background design is the methodological bottleneck. Figure 9 illustrates this interpretation bridge.

4.6 Limitations

1. Pseudo-absences remain inferred labels.

2. Road accessibility is an imperfect survey-effort proxy. Moreover, road distance plays a dual role in our design: it is the highest tautology-correlated proxy ($|\rho| = 0.307$

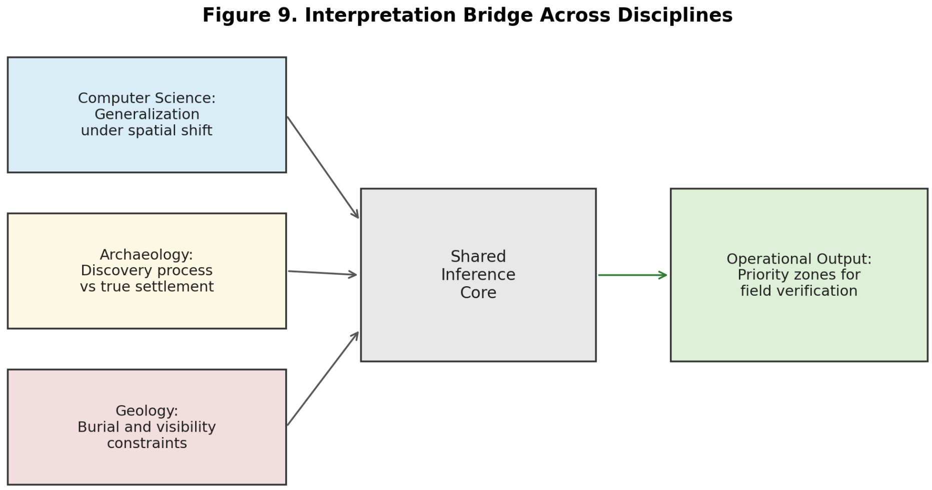

Figure 9: Interpretation bridge showing how CS, archaeology, and geology converge on operational survey prioritisation.

in Test 1) *and* the basis for TGB pseudo-absence selection. A reviewer may ask whether TGB gains (E010–E013) partly reflect the model learning road-accessibility patterns through pseudo-absence composition rather than genuine environmental signal. Three observations mitigate this concern: (a) road distance is *not* included as a training feature in the final E013 configuration; (b) Test 3 shows that model performance *improves* in the least road-accessible quartile (Q4 AUC $= 0.788 >$ Q1 AUC $= 0.731$), the opposite of what road-driven learning would produce; and (c) E013 exceeds the DKNS spatial-interpolation ceiling by $+0.122$ AUC using environmental features alone. Nevertheless, this coupling between sampling design and potential confound deserves explicit attention in future work, ideally through comparison with non-road-based survey-effort proxies.

3. Hard-negative realisation can exceed nominal targets.

4. External transfer beyond East Java is untested.

5. Block-size sensitivity was tested only at three coarse scales.

5 Conclusions

E013 is the first configuration to clear the operational threshold (seed-averaged AUC 0.751, best single-run 0.768), while robustness estimates remain near-threshold but stable. The core methodological conclusion is that, under survey-biased archaeological data, pseudo-absence realism is the dominant determinant of spatial transfer. We acknowledge two important caveats: (1) the seed-averaged AUC of 0.751 is only marginally above the 0.75 MVR threshold, and 9 of 20 seeds fall below it, indicating that the model operates near its performance boundary; and (2) the tautology test suite yields a conditional rather than unconditional pass, with T1–T2 metrics in the grey zone. These near-threshold results are honest reflections of the difficulty of the problem rather than grounds for rejection — the model consistently exceeds null-model baselines and generalises to held-out sites in temporal validation. This provides an actionable, interdisciplinary baseline for target prioritisation and subsequent field validation.

Supplementary Materials

The following supplementary materials are available online:

- **Table S1:** Seed stability analysis — XGBoost and RandomForest AUC across 20 alternate random seeds for E013.
- **Table S2:** Block-size sensitivity — spatial CV performance at ˜40, ˜50, and ˜60 km block sizes.
- **Figure S1:** Full feature importance comparison across E007–E013 for both algorithms.

Acknowledgements

The authors thank reviewers for feedback on model framing, validation, and interpretation.

Funding

This research received no external funding.

Competing Interests

The authors declare no competing interests.

Authors' Contributions

M.A. conceived the research design, developed the methodology, implemented software, conducted formal analysis, curated data, created visualisations, and wrote the original draft. G.F.G. contributed to methodology refinement, code review, and manuscript editing.

Data Availability

All data used in this study are available as follows:

Archaeological site data: The compiled East Java archaeological site dataset (666 sites) is available in the repository at `data/processed/east_java_sites.geojson`. Sources include OpenStreetMap (ODbL licence), Wikidata (CC BY-SA), Wikipedia Indonesia, and BPCB Jawa Timur publications.

Environmental covariates: The Copernicus GLO-30 DEM and derived layers (slope, aspect, TWI, TRI, river distance, road distance) are available from the Copernicus Programme (ESA) at `https://dataspace.copernicus.eu/` (free and open licence). OpenStreetMap road and river data are available from `https://openstreetmap.org` (ODbL licence).

Interactive dashboard: An interactive dashboard for exploring model predictions, SHAP analysis, zone classification, and validation results is available at `https://volcarch-dashboard.streamlit.app`.

Code and software: All source code, including the E013 hybrid model implementation, temporal validation (E014), and robustness checks, is available at GitHub:

https://github.com/neimasilk/volcarch-repo (MIT licence).

Dependency lock: Full Python package versions are documented in requirements_submission_lo

AI Disclosure

AI-assisted tools (Claude, Anthropic) were used during manuscript preparation for code development assistance, data analysis scripting, and figure generation. The AI tools did not contribute to research design, hypothesis formulation, or scientific interpretation. All results were independently verified by the authors.

References

Omri Allouche, Asaf Tsoar, and Ronen Kadmon. Assessing the accuracy of species distribution models: prevalence, kappa and the true skill statistic (TSS). *Journal of Applied Ecology*, 43(6):1223–1232, 2006. doi: 10.1111/j.1365-2664.2006.01214.x.

Miguel B. Araújo and Antoine Guisan. Five (or so) challenges for species distribution modelling. *Journal of Biogeography*, 33(10):1677–1688, 2006. doi: 10.1111/j.1365-2699.2006.01584.x.

Miguel B. Araújo and Mark New. Ensemble forecasting of species distributions. *Trends in Ecology & Evolution*, 20(11):609–613, 2005. doi: 10.1016/j.tree.2005.09.010.

Morgane Barbet-Massin, Frédéric Jiguet, Cécile H. Albert, and Wilfried Thuiller. Selecting pseudo-absences for species distribution models: how, where and how many? *Methods in Ecology and Evolution*, 3(2):327–338, 2012. doi: 10.1111/j.2041-210X.2011.00172.x.

Jean-Louis Bourdier, Indyo Pratomo, Jean-Claude Thouret, Georges Boudon, and P. M. Vincent. Observations, stratigraphy and eruptive processes of the 1990 eruption of Kelut volcano, Indonesia. *Journal of Volcanology and Geothermal Research*, 79(3–4):181–203, 1997. doi: 10.1016/S0377-0273(97)00031-0.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324.

Marco Castiello and Marj Tonini. Machine learning applications in archaeological predictive modelling. *Journal of Computer Applications in Archaeology*, 4(1):110–125, 2021. doi: 10.5334/jcaa.71.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. ACM, 2016. doi: 10.1145/2939672.2939785.

Douglas C. Comer, Michael J. Harrower, Joy McCorriston, Benjamin O'Meara, and Karim Sadr. Sampling methods for archaeological predictive modeling. *Journal of Archaeological Science: Reports*, 47:103824, 2023. doi: 10.1016/j.jasrep.2022.103824.

D. J. De Groot. Volcanic hazards in the archaeological record of Mount Merapi, Central Java, Indonesia. *Archaeology in Oceania*, 44(3):131–145, 2009. doi: 10.1002/j.1834-4453.2009.tb00045.x.

Jane Elith, Catherine H. Graham, Robert P. Anderson, Miroslav Dudík, Simon Ferrier, Antoine Guisan, Robert J. Hijmans, Falk Huettmann, John R. Leathwick, Anthony Lehmann, Jin Li, Lucia G. Lohmann, Bette A. Loiselle, Glenn Manion, Craig Moritz, Miguel Nakamura, Yoshinori Nakazawa, Jacob McC. Overton, A. Townsend Peterson, Steven J. Phillips, Karen Richardson, Ricardo Scachetti-Pereira, Robert E. Schapire, Jorge Soberón, Stephen Williams, Mary S. Wisz, and Niklaus E. Zimmermann. Novel methods improve prediction of species' distributions from occurrence data. *Ecography*, 29(2):129–151, 2006. doi: 10.1111/j.2006.0906-7590.04596.x.

Arnau Folch. A review of tephra transport and dispersal models: evolution, current status, and future perspectives. *Journal of Volcanology and Geothermal Research*, 235–236:96–115, 2012. doi: 10.1016/j.jvolgeores.2012.05.020.

Charles French, Nick Whalen, Cathy Batt, Andrew Lovett, and Neil Oakey. Monitoring archaeological site burial in England: The assessment of topsoil thickness at selected Roman villa sites. *Antiquity*, 77(295):116–130, 2003. doi: 10.1017/S0003598X0006137X.

Stefanos Georganos, Tais Grippa, Assane Niang Gadiaga, Catherine Linard, Moritz Lennert, Sabine Vanhuysse, Nicholus Brücker, and Eléonore Wolff. Geographical random forests: a spatial extension of the random forest algorithm to address spatial heterogeneity in remote sensing and population modelling. *Geocarto International*, 36 (2):121–136, 2021. doi: 10.1080/10106049.2019.1595177.

John Grattan and Robin Torrence. Volcanism, archaeology and the cargoes of commerce. *Quaternary International*, 162–163:109–111, 2007. doi: 10.1016/j.quaint.2006.10.017.

Alberto Jiménez-Valverde, Narayani Barve, Andrés Lira-Noriega, Sean P. Maher, Yoshinori Nakazawa, Monica Papeş, Jorge Soberón, and A. Townsend Peterson. Defining the fundamental niche: Implications for species distribution modelling. *Journal of Biogeography*, 47(12):2529–2541, 2020. doi: 10.1111/jbi.13968.

Hans Kamermans, Wieke de Neef, Jan Francke Porck, and Jitte Waagen. Archaeological predictive modelling in the Netherlands: a quantitative approach to heritage management. *Archaeological Prospection*, 26(4):295–306, 2019. doi: 10.1002/arp.1761.

Ron Kohavi. A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Proceedings of the 14th International Joint Conference on Artificial Intelligence*, volume 2, pages 1137–1143, Montreal, Quebec, Canada, 1995. Morgan Kaufmann Publishers Inc.

Kenneth L. Kvamme. There and back again: Revisiting archaeological locational modeling. In Mark W. Mehrer and Konnie L. Wescott, editors, *GIS and Archaeological Site Location Modeling*, pages 3–38. Taylor & Francis, Boca Raton, 2006.

Jorge M. Lobo, Alberto Jiménez-Valverde, and Joaquín Hortal. The uncertain nature of absences and their importance in species distribution modelling. *Ecography*, 33(1): 103–114, 2010. doi: 10.1111/j.1600-0587.2009.06039.x.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, pages 4765–4774. Curran Associates, Inc., 2017.

Larry G. Mastin, Alexa R. Van Eaton, and Jacob B. Lowenstern. Modeling ash fall distribution from a Yellowstone supereruption. *Geochemistry, Geophysics, Geosystems*, 15(8):3459–3475, 2014. doi: 10.1002/2014GC005469.

Larry G. Mastin, Alexa R. Van Eaton, David J. Schneider, Kristi L. Wallace, Donald A. Swanson, Charles R. Bacon, Manuel Nathenson, Richard B. Waittt, Cynthia A. Gardner, Alison B. Till, et al. Lessons for forecasting eruptions at Yellowstone and other large volcanoes from the 1980 eruption of Mount St. Helens. *Bulletin of Volcanology*, 84(7):78, 2022. doi: 10.1007/s00445-022-01613-0.

Christopher G. Newhall and Stephen Self. The Volcanic Explosivity Index (VEI): An estimate of explosive magnitude for historical volcanism. *Journal of Geophysical Research: Oceans*, 87(C2):1231–1238, 1982. doi: 10.1029/JC087iC02p01231.

Laure Nuninger, Philip Verhagen, François Bertoncello, and Angelo Castrorao Barba. Estimating "land use heritage" to model changes in archaeological settlement patterns. In Juan A. Barcelo and Igor Bogdanovic, editors, *Mathematics and Archaeology*, pages 223–238. CRC Press, Boca Raton, 2016.

Steven J. Phillips, Robert P. Anderson, and Robert E. Schapire. Maximum entropy modeling of species geographic distributions. *Ecological Modelling*, 190(3–4):231–259, 2006. doi: 10.1016/j.ecolmodel.2005.03.026.

Steven J. Phillips, Miroslav Dudík, Jane Elith, Catherine H. Graham, Anthony Lehmann, John Leathwick, and Simon Ferrier. Sample selection bias and presence-only distribution models: implications for background and pseudo-absence data. *Ecological Applications*, 19(1):181–197, 2009. doi: 10.1890/07-2153.1.

David M. Pyle. The thickness, volume and grainsize of tephra fall deposits. *Bulletin of Volcanology*, 51(1):1–15, 1989. doi: 10.1007/BF01086757.

David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J. Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, David I. Warton, Brendan A. Wintle, Florian Hartig, and Carsten F. Dormann. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017. doi: 10.1111/ecog.02881.

Sabrina D. Senay, Susan P. Worner, and Takayoshi Ikeda. Novel three-step pseudo-absence selection technique for improved species distribution modelling. *PLOS ONE*, 8(8):e71218, 2013. doi: 10.1371/journal.pone.0071218.

Lee Siebert, Tom Simkin, and Paul Kimberly. *Volcanoes of the World*. Smithsonian Institution / University of California Press, Berkeley, 3rd edition, 2010.

Robin Torrence, Trudy Doelman, Stephen Cotter, Rachel Popelka-Filcoff, Cosmas Pavlides, John Webb, and Ron Richards. Pleistocene colonisation of the Bismarck Archipelago: New evidence from West New Britain. *Archaeology in Oceania*, 37(3): 145–151, 2002. doi: 10.1002/j.1834-4453.2002.tb00517.x.

Roozbeh Valavi, Jane Elith, José J. Lahoz-Monfort, and Gurutzeta Guillera-Arroita. blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution*, 10(2):225–232, 2019. doi: 10.1111/2041-210X.13107.

Philip Verhagen and Thomas G. Whitley. Integrating archaeological theory and predictive modeling: a live report from the scene. *Journal of Archaeological Method and Theory*, 19(1):49–100, 2012. doi: 10.1007/s10816-011-9102-7.

Philip Verhagen, Laure Nuninger, François Bertoncello, and Rachel Opitz. Whither predictive modelling? A panel discussion. *Archaeological and Anthropological Sciences*, 12 (1):14, 2020. doi: 10.1007/s12520-019-00937-1.

LuAnn Wandsnider. The spatial dimension of time-averaging in archaeological assemblages. *Bulletin of the Society for American Archaeology*, 10(4):17–19, 1992.

Cheng Wang, Ying Li, Yijian Liu, Huan Liu, Jian Liu, Yijian Su, and Huadong Guo. A machine-learning approach to geospatial predictive modelling in archaeology. *ISPRS International Journal of Geo-Information*, 12(6):238, 2023. doi: 10.3390/ijgi12060238.

Mary S. Wisz and Antoine Guisan. Do pseudo-absence selection strategies influence species distribution models and their predictions? An information-theoretic approach based on simulated data. *BMC Ecology*, 9:8, 2009. doi: 10.1186/1472-6785-9-8.